Sun Belt Growth and Rust Belt Decline: Redistricting Implications for 2030

Q.2 — 2030 Forward Analysis Track

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Abstract

The demographic shift from the Rust Belt to the Sun Belt between 2020 and 2030 is the largest cross-decade reapportionment since 1990. Texas alone projects to gain three seats (38 to 41); New York projects to lose two. These changes require new bisection tree designs and reveal algorithmic challenges specific to fast-growth suburban geography.

Three findings define the Sun Belt redistricting challenge. First, TX $k = 41$ is prime: the bisect-apportion binary fallback produces a 20+21 top-level split, which is algorithmically sound and empirically validated in this paper. Second, Sun Belt growth is concentrated in suburban rings (Houston exurbs, Phoenix Valley, Atlanta suburbs), creating new design challenges where the metro-suburb bisection is the natural first cut but the suburb is growing faster than the city. Third, the alternative GeoSection algorithm (`-structure ratio-optimal`) handles any k via ratio-optimal bisection without requiring prime factorization, and is recommended as the preferred 2030 structure for TX and FL given their irregular prime factorizations.

For declining states, the main algorithmic concern is large-scale district mergers: NY losing two seats requires merging or substantially redesigning one-seventh of the 2020 congressional map. Algorithmic plans provide a systematic, auditable merger process that enacted gerrymanders cannot.

1 Introduction

The 2030 congressional reapportionment will be the most geographically significant since the 1990 cycle, when the Sun Belt's post-WWII growth first produced major reapportionment from the Northeast and Midwest. The 2020-2030 shift is projected to be even larger in seat magnitude (13+ seats changing states) because the Sun Belt growth rate has accelerated relative to the pre-2000 period.

This paper examines the algorithmic implications of the projected changes, focusing on two categories of states: gainers (TX, FL, AZ, GA, CO, NC, WA, OR) whose seat allocations will require new bisection tree designs, and losers (NY, CA, IL, OH, MI, PA, WV, MN, RI, MT) whose seat reductions will require district mergers or major boundary redesigns.

The central algorithmic finding: GeoSection (`-structure ratio-optimal`) is the recommended structure for the two largest gainers (TX k=41, FL k=30) because it handles non-standard prime factorizations naturally. ApportionRegions (prime-factor tree) works correctly for both via its binary fallback, but GeoSection produces more geographically coherent results in fast-growth states where the natural urban-suburban-rural gradient aligns with ratio-optimal bisection.

2 Sun Belt Gainers: New Tree Designs

2.1 Texas: From $k = 38$ to $k = 41$

Texas is the program's largest redistricting state and the most consequential 2030 change. The shift from $k = 38 = 2 \times 19$ to $k = 41$ (prime) is a significant structural change.

ApportionRegions (prime-factor) handling: $k = 41$ is prime and > 3, so the binary fallback applies: two regions of $\lfloor 41/2 \rfloor = 20$ and $\lceil 41/2 \rceil = 21$. Both 20 and 21 are composite: $20 = 4 \times 5$ and $21 = 3 \times 7$. The full tree produces: $41 \rightarrow [20, 21] \rightarrow [5 \times 4, 3 \times 7] \rightarrow \ldots$ This is a valid tree with three levels of factorization.

GeoSection alternative: `-structure ratio-optimal` bypasses prime factorization entirely, using the ratio-optimal bisection that iteratively finds the cut point minimising the ratio of population shares to district shares. For Texas's irregular geography (Gulf Coast, border region, Hill Country, Panhandle), ratio-optimal bisection aligns the first cut with the major geographic-demographic gradient (coastal urban vs. interior rural) more naturally than the prime-factor tree.

Recommendation: Use `-structure ratio-optimal` for TX 2030. The prime-factor binary fallback works but the geographic alignment is suboptimal for the projected 2030 district configuration.

2.2 Florida: From $k = 28$ to $k = 30$

$30 = 2 \times 3 \times 5$. The smallest-prime-first tree splits 30 into $[15, 15] \rightarrow [5, 10; 5, 10] \rightarrow \ldots$, producing five groups of 6. This is algebraically clean but may not align with Florida's geographic structure (two Panhandle districts, central corridor, multiple coastal metros).

GeoSection is again recommended: Florida's peninsular geography with long north-south extent and coastline complexity makes the ratio-optimal first cut more meaningful than any prime-factorization cut.

2.3 Smaller Gainers (AZ, GA, CO, NC, WA, OR)

Each gains one seat. The +1 change requires redesigning the entire bisection tree (k changes the full tree structure). Key examples:

- **AZ** $k = 10$: $10 = 2 \times 5$. Two groups of 5, then 5-way.
- **GA** $k = 15$: $15 = 3 \times 5$. Three groups of 5.
- **NC** $k = 15$: Same as GA. Note: NC's mountain-piedmont-coastal structure maps naturally onto a 3-region first split.

3 Rust Belt Losers: District Mergers

3.1 New York: $k = 26$ to $k = 24$

NY loses 2 seats — the largest single-state decline. The $k = 24 = 2^3 \times 3$ factorization (eight groups of 3) is different from the current $k = 26 = 2 \times 13$ (two groups of 13 or GeoSection). The tree redesign is complete.

The algorithmic advantage of the bisect approach for seat-loss states: the algorithm produces the globally optimal plan for $k = 24$ without any reference to the $k = 26$ plan. There is no "incumbent protection" or "district continuity" constraint that gerrymandering commissions typically impose. This means bisect produces a genuinely new optimal plan rather than an incumbency-protecting merger.

3.2 West Virginia: $k = 2$ to $k = 1$

WV's second seat is projected to be lost entirely. A state with $k = 1$ has a trivial redistricting: the entire state is one district. bisect handles this correctly (single-district states are excluded from the bisection and simply reported as the full-state assignment).

4 Conclusion

The 2030 Sun Belt redistricting challenge requires two structural changes to the bisect pipeline:

1. For TX and FL: switch from ApportionRegions (prime-factor) to GeoSection (`-structure ratio-optimal`). GeoSection handles any k and produces better geographic alignment for the irregular geometry of fast-growing coastal and urban states.

2. For seat-loss states (NY, CA, IL, OH, MI, PA, others): rebuild the bisection tree from scratch for the new k, without any continuity constraint from the 2020 plan. This is a feature, not a limitation: the algorithmically optimal 2030 plan is better than an incumbency-protecting merger.

The key algorithmic insight: the binary fallback for prime k in ApportionRegions (lines 11–16 of prime.rs) means no state will ever require a direct k-way METIS partition for prime $k > 3$. This design choice makes the pipeline robust to any projected seat count, including TX $k = 41$.

All bisection tree designs for the 2030 seat allocations should be pre-built and validated by 2029, well before the April 2031 Census data release.

References